

Discuss and recall — fractions

What Grade 5 established about fractions, gathered on one page before the division begins.

LESSON 21

1 × 40 MIN

MATEMATIKA 6, TAKRORLASH

S7 7 ENTRY

LESSON CLOCK

8'

10'

12'

10'

The vocabulary	8 min
Simplifying and converting	10 min
Comparing	12 min
The three operations you already have	10 min

LEARNING OBJECTIVES

- Simplify a fraction to its lowest terms.
- Convert between mixed numbers and improper fractions.
- Compare and order fractions.
- Add, subtract and multiply fractions.

TEXTBOOK

National	pp. 54–57
Cambridge	Stage 7, pp. 66–70
Workbook	Exercise 7.1

01

Explanation

The vocabulary

KIND	EXAMPLE	TEST
proper	$\frac{3}{5}$	numerator smaller
improper	$\frac{7}{5}$	numerator larger
mixed	$1\frac{2}{5}$	a whole part and a proper fraction
equal to one	$\frac{5}{5}$	the two are equal

A FRACTION IS A DIVISION WAITING TO HAPPEN

$\frac{7}{5}$ means $7 \div 5$, which is 1 remainder 2 — that is, $1\frac{2}{5}$. The two forms are the same number written for different purposes.

Simplifying and converting

TASK	METHOD	EXAMPLE
simplify	divide both parts by their HCF	$\frac{18}{24} = \frac{3}{4}$
mixed to improper	whole $\times$ denominator, plus numerator	$2\frac{3}{4} = \frac{11}{4}$
improper to mixed	divide, keep the remainder	$\frac{17}{5} = 3\frac{2}{5}$
equivalent fractions	multiply both parts by the same number	$\frac{2}{3} = \frac{8}{12}$

SIMPLIFY AT THE END, ALWAYS

An answer of $\frac{12}{18}$ is not wrong but is not finished; $\frac{2}{3}$ is. Marks for “lowest terms” are lost far more often than marks for the arithmetic itself.

Comparing

PAIR	METHOD	RESULT
$\frac{3}{7}$ and $\frac{5}{7}$	same denominator	$\frac{3}{7} < \frac{5}{7}$
$\frac{3}{5}$ and $\frac{3}{8}$	same numerator — bigger denominator, smaller parts	$\frac{3}{8} < \frac{3}{5}$
$\frac{3}{5}$ and $\frac{5}{8}$	common denominator 40	$\frac{24}{40} < \frac{25}{40}$
$\frac{4}{9}$ and $\frac{1}{2}$	compare with $\frac{1}{2}$	$\frac{4}{9} < \frac{1}{2}$

BIGGER DENOMINATOR, SMALLER PIECES

A cake cut into 8 gives smaller slices than the same cake cut into 5. That single picture settles every comparison with equal numerators.

The three operations you already have

$$\frac{a}{b} + \frac{c}{b} = \frac{a + c}{b}$$
$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

TASK	WORKING	ANSWER
$\frac{2}{3} + \frac{1}{4}$	$\frac{8}{12} + \frac{3}{12}$	$\frac{11}{12}$
$\frac{5}{6} - \frac{1}{3}$	$\frac{5}{6} - \frac{2}{6}$	$\frac{1}{2}$
$\frac{2}{3} \cdot \frac{3}{5}$	cancel the 3s	$\frac{2}{5}$
$\frac{3}{4}$ of 20	$20 \div 4 \cdot 3$	15

ONE OPERATION IS STILL MISSING

Adding, subtracting and multiplying are done. Division is the whole of the next four lessons — and it turns out to be the easiest of the four.

02 Worked examples

EXAMPLE 1 Simplify $\frac{18}{24}$ and convert $\frac{17}{5}$ to a mixed number.

- ① HCF of 18 and 24 is 6.

$$\frac{3}{4}$$

- ② $17 \div 5 = 3$ remainder 2.

$$3\frac{2}{5}$$

ANSWER

$$\frac{3}{4} \text{ and } 3\frac{2}{5}$$

EXAMPLE 2 Which is larger, $\frac{3}{5}$ or $\frac{5}{8}$?

- ① Common denominator 40.

② $\frac{3}{5} = \frac{24}{40}$

③ $\frac{5}{8} = \frac{25}{40}$

- ④ $\frac{5}{8}$ is larger.

ANSWER

$$\frac{5}{8}$$

EXAMPLE 3 Compute $\frac{2}{3} \cdot \frac{3}{5}$.

① Cancel the 3 in the numerator against the 3 in the denominator.

② $\frac{2}{5}$

Cancel before multiplying — it is always quicker.

ANSWER

$$\frac{2}{5}$$

03 Interactive model

Ask which is bigger, a fifth or an eighth, before anything else; the answer that comes back tells you exactly how much recall this class needs.

Simplifying to lowest terms

$$\frac{18}{24}$$

① $18 = 2 \cdot 3^2$

, $24 = 2^3 \cdot 3$

Factorise
both.

② The HCF is

6 .

The
highest
one
—
anything
less
leaves
work.

③ $\frac{3}{4}$

Divide
both
parts
by
it.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Proper fraction	To'g'ri kasr	Правильная дробь
Improper fraction	Noto'g'ri kasr	Неправильная дробь
Mixed number	Aralash son	Смешанное число
Lowest terms	Qisqarmas ko'rinish	Несократимый вид
Common denominator	Umumiy maxraj	Общий знаменатель
Reciprocal	Teskari son	Обратное число
To simplify	Qisqartirish	Сократить
To compare	Taqqoslash	Сравнить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

7 $\frac{5}{4}$ is:

proper

improper

mixed

equal to one

2 $\frac{3}{4}$ as an improper fraction:

$$\frac{9}{4}$$

$$\frac{11}{4}$$

$$\frac{5}{4}$$

$$\frac{23}{4}$$

18 24 simplifies to:

$$\frac{9}{12}$$

$$\frac{3}{4}$$

$$\frac{2}{3}$$

$$\frac{6}{8}$$

Which is larger, 3 5 or 3 8 ?

$$\frac{3}{5}$$

$$\frac{3}{8}$$

equal

cannot say

2 3 + 1 4 equals:

$$\frac{3}{7}$$

$$\frac{11}{12}$$

$$\frac{2}{12}$$

$$\frac{3}{12}$$

3 4 of 20 is:

5

15

16

60

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $\frac{18}{24}$ simplified

ANSWER $\frac{3}{4}$

2. $\frac{45}{60}$ simplified

ANSWER $\frac{3}{4}$

3. $2\frac{3}{4}$ as improper

ANSWER $\frac{11}{4}$

4. $\frac{17}{5}$ as mixed

ANSWER $3\frac{2}{5}$

5. $\frac{3}{7}$ or $\frac{5}{7}$: larger

ANSWER $\frac{5}{7}$

6. $\frac{3}{5}$ or $\frac{3}{8}$: larger

ANSWER $\frac{3}{5}$

7. $\frac{2}{3} + \frac{1}{4}$

ANSWER $\frac{11}{12}$

Medium · the standard question

7 PROBLEMS

1. $\frac{5}{6} - \frac{1}{3}$

ANSWER $\frac{1}{2}$

2. $\frac{2}{3} \cdot \frac{3}{5}$

ANSWER $\frac{2}{5}$

3. $\frac{3}{4}$ of 20

ANSWER 15

4. $\frac{3}{5}$ or $\frac{5}{8}$: larger

ANSWER $\frac{5}{8}$

5. $\frac{4}{9}$ or $\frac{1}{2}$: larger

ANSWER $\frac{1}{2}$

6. $1\frac{1}{2} + \frac{3}{4}$

ANSWER $2\frac{1}{4}$

7. $\frac{2}{5} \cdot \frac{10}{3}$

ANSWER $1\frac{1}{3}$

Hard · stretch and reason

7 PROBLEMS

1. Order $\frac{2}{3}, \frac{3}{5}, \frac{5}{8}$

ANSWER $\frac{3}{5} < \frac{5}{8} < \frac{2}{3}$

2. $2\frac{1}{3} - 1\frac{1}{2}$

ANSWER $\frac{5}{6}$

3. $\frac{5}{8}$ of 64

ANSWER 40

4. A fraction equal to $\frac{3}{4}$ with denominator 28

ANSWER $\frac{21}{28}$

5. $\frac{3}{4} \cdot \frac{2}{9}$

ANSWER $\frac{1}{6}$

6. Which is nearer to 1, $\frac{7}{8}$ or $\frac{9}{10}$?

ANSWER $\frac{9}{10}$

7. Why is $\frac{12}{18}$ an unfinished answer?

ANSWER Both parts still divide by 6

07 Homework

Homework — 5 tasks

Every answer in lowest terms, and mixed numbers where the fraction is improper.

1. Simplify $\frac{24}{36}$, $\frac{30}{45}$ and $\frac{16}{40}$.
2. Convert $3\frac{2}{5}$ to an improper fraction and $\frac{23}{6}$ to a mixed number.
3. Order $\frac{1}{2}$, $\frac{3}{7}$, $\frac{5}{9}$.
4. Compute $\frac{3}{4} + \frac{2}{5}$ and $\frac{7}{8} - \frac{1}{4}$.
5. Compute $\frac{3}{5} \cdot \frac{10}{9}$, cancelling first.